

Lily Zhang

San Francisco Bay Area · lilyzhng.ai@gmail.com · lilyzh.ng · in/lilyzhng · @lily_gpupoor · github.com/lilyzhng · [scholar](#)

CAREER STATEMENT

Applied research lead with 8 years of experience building AI products for post-training, including generating training data, designing evaluations, and creating agentic post-training pipelines for researchers and ML engineers. My multimodal data agents serve 76+ ML engineers daily for data search and curation at a 95% auto-acceptance rate. I built a QA bench with an evolving failure taxonomy that catches agent behavior degradation before any new release reaches the customer. My post-trained AI infra agents run GPU demand forecasting in production at a 91% adoption rate.

HIGHLIGHTS

- **Ran LLM Post-Training Release Cycles (Eureka, Qwen3-32B):** Worked side-by-side with ML researchers to post-train an LLM agent that generates feature code for GPU demand forecasting, with over 300 training runs: 2 stage trainings, first SFT on curated domain Q&A for cold-start knowledge; the second cycle, then RL (GRPO) with a reward design that prevents reward hacking. Delivered production release with 91% production adoption, +16% demand fulfillment, and 33% reduction in GPU compute migration. [NeurIPS-W 2025](#) · [DASFAA 2026 Oral](#).
- **Curated Evolving Evaluation Data for Agentic Workflows:** Defined the org's first failure taxonomy for the data-QA agent and built a multimodal benchmark (320 tasks) around it, gating every agent release by catching regressions before production. The taxonomy made agent performance measurable per failure mode, which enabled routing orchestration: a lightweight orchestration instead of calling the VLM every time, cutting inference cost by 90%.
- **Anticipated Evaluation Needs Across Safety Release Cycles:** The safety guard model carried two safety risks: over-refusal (safe queries misclassified as unsafe, blocking users) and under-refusal (unsafe queries let through). RL post-trained for six release cycles, designed a constitution board with amendment governance and C-LIM, a metric calibrating every cell for learnable impact, lifting balanced accuracy from 0.876 to 0.921, reducing both over-refusal and under-refusal safety risks. [C-Guard](#) (COLM 2026 Efficient Reasoning).
- **Built LosslessBench, Speculative Decoding Evaluation:** Curated evaluation data to gauge whether model behavior survives speculative decoding post-training, across frontend design, creative writing, guardrails, coding, and agent workflows. Used as a behavior-regression gate rather than a training signal: it caught a model degradation in the frontend-design task. Companion teaching material submitted to the NeurIPS 2026 Education Track. [Blogpost](#) · [Dataset](#)

WORK EXPERIENCE

Latitude AI · Technical Lead, AI & ML · Feb 2024 – Present

- **Ran Model Release Cycles with Evaluation:** Released the 50,000-model world-simulation campaign in evaluation-gated cycles: a 2K pilot judged on perceptual quality, reconstruction accuracy, and temporal consistency; a 20K run confirming the metrics held at scale; then the 50K production release (300K GPU hours in two months) generating synthetic video at 92.5% lower cost per sequence.
- **Launched Alfa Search, the First VLM Product:** Owned the org's first VLM workstream, built the eval suite with targeted synthetic data augmentation, won leadership buy-in and compute investment, and launched a text-video data search and curation agent, used daily by 76+ ML engineers at a 95% auto acceptance rate.
- **Investigated and Resolved Critical Training Issues:** Root-caused a stalled org-wide training run within 24 hours across time zones, confirming a missing cache overloading HPC file counts, and worked closely with ML researchers to resolve it. Managed three ML engineers to set up the entire training pipeline within one quarter, resulting in an 8.1% F1 improvement over baseline.
- **Measured Data Quality Trade-Offs:** Drove 200+ training and evaluation experiments with researchers to a data-mix decision (pretrain synthetic, finetune real).

Ford Greenfield Labs · Technical Project Lead, AI & ML · Jun 2022 – Jan 2024

- **Bridged Global Teams to Drive Action:** Coordinated a 10+ engineer data-generation program across the US, India, Germany, and Israel with no reporting authority, setting the product roadmaps and running PI planning. Presented results in Ford leadership reviews.
- **Synthetic Data Vendor Evaluation:** Buyer-side owner of vendor sourcing and evaluation across 10+ vendors (Scale AI, Parallel Domain, Applied Intuition, NVIDIA); gated major purchases with pilot-batch POCs.
- **Shipped the Data Engine Behind 492K Vehicles:** Delivered the Continuous Learning Data Engine for Ford BlueCruise offline perception. The first 15K sequences saved \$1.1M in labeling.

Ford Greenfield Labs · Sr. Research Scientist · Jun 2019 – May 2022

- **Shipped Research as Product Features:** Turned synthetic-data research into Pro Trailer Hitch Assist on F-150, Lightning, and Super Duty. Henry Ford Technology Award, top <0.1% of ~177K employees.

DEVELOPER PRODUCT LAUNCH

- **SofaGenius** (2026): agentic post-training platform that curates data, launches GPU runs, and monitors anomaly. Adopted for internal ML experiments. Anthropic hackathon finalist, top 15/13k. [Project](#) · [Video](#)
- **Deep-Seek:** deep-research agent, LLM-powered retrieval compiling structured entities from internet-scale sources. Maintainer, 510+ stars, adopted by users incl. Microsoft, Amazon, and Bloomberg. [Repo](#)
- **Agent Channels** (2026): talk to CLI coding agents from Discord. [Repo](#). **Agent in Chrome** (2026) [Repo](#)
- **Frontend Slides** (2026): contributor who drove the system multimodal, where image content drives slide composition. 25K+ GitHub stars. [Repo](#)

PUBLICATIONS

- [LosslessBench](#) (2026): cross-provider evaluation of inference acceleration's behavioral cost, built before speed-vs-quality became the public debate. Its 600 tasks later settled a viral vendor dispute. [Dataset](#)
- **C-Guard: A Constitution-Grid Instrument for Data-Efficient RL Alignment** (COLM 2026 Efficient Reasoning). [Paper](#) · [Blogpost](#)
- 9 research papers (CVPR, ICCV, WACV, NeurIPS, COLM, RA-L) and 7 granted patents. [scholar](#)

TECHNICAL LEADERSHIP & COMMUNICATION

- **Research Advisor, AliCloud (2025):** Mentored research interns and guided technical direction for Eureka's LLM post-training research through to production (see Highlights).
- **General Chair, RoboGen Workshop, IEEE IROS 2025** (world generation for robot learning).
- **Keynote:** "Generative AI: A New World Simulator for Synthetic Data," GenAI Summit 2024.
- **Keynote:** "Synthetic Data Diversification," NVIDIA GTC 2023 · Panelist, AI/ML infra, NVIDIA GTC 2026.
- **Keynote:** "From Lab to Road: Driving Autonomous Driving Research to Production," IEEE IROS 2023.

SKILLS

- **Product Management:** 0-to-1 product launch, cross-team alignment and executive communication
- **Evals:** evaluation suite design, benchmark curation, failure taxonomy, regression investigation, LLM/VLM-as-judge, human-in-the-loop systems
- **Post-Training:** SFT, GRPO, reward design, checkpoint analysis, convergence monitoring
- **Technical:** Python, SQL, PyTorch, Ray, large-scale training

EDUCATION

- Stanford University · Graduate Program, AI, School of Engineering (2022–2024)
- Penn State University · B.S. Computer Science (2013–2017)